

JKBOSE

2026 CLASS 11TH

MOTION IN PLANE

LECTURE-1

PHYSICS

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Topics To Be Covered

- 1** Revision
- 2** Motion in 1D (Straight Line Motion)
- 3** Motion in 2D (Motion in a Plane)
- 4** Motion in 3D (Motion in a Plane)
- 5** Motion in 2D ✔

Topics To Be Covered

- 6** Projectile Motion
- 7** Equation of Trajectory
- 8** ..

“Mehnat Karte Rahoo”

Recap from 1st Lecture

Vector

- The pd have magnitude and direction



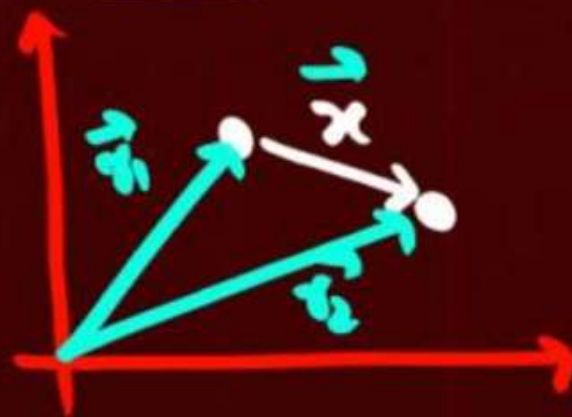
A
→
A

Types of Vector

(i) Position vector



(ii) Displacement V



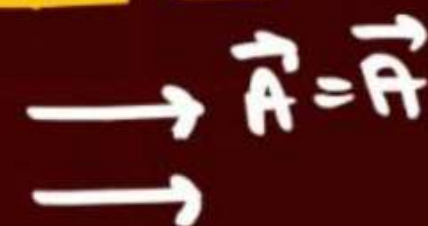
(iii) Parallel vector



(iii) Antiparallel vector



(iv) Equal vector



(v) Orthogonal vector



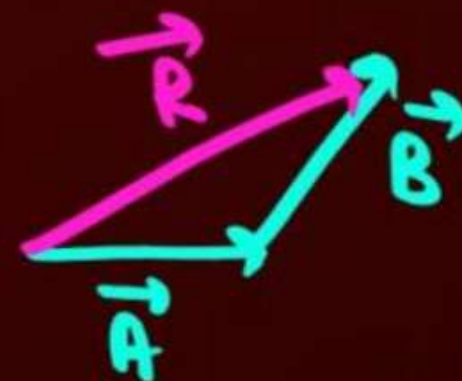
(vi) Unit vector

$n=1$
Direction



+ - × ÷
A - B
C B
r a
vector

① Triangular law of vector Addition

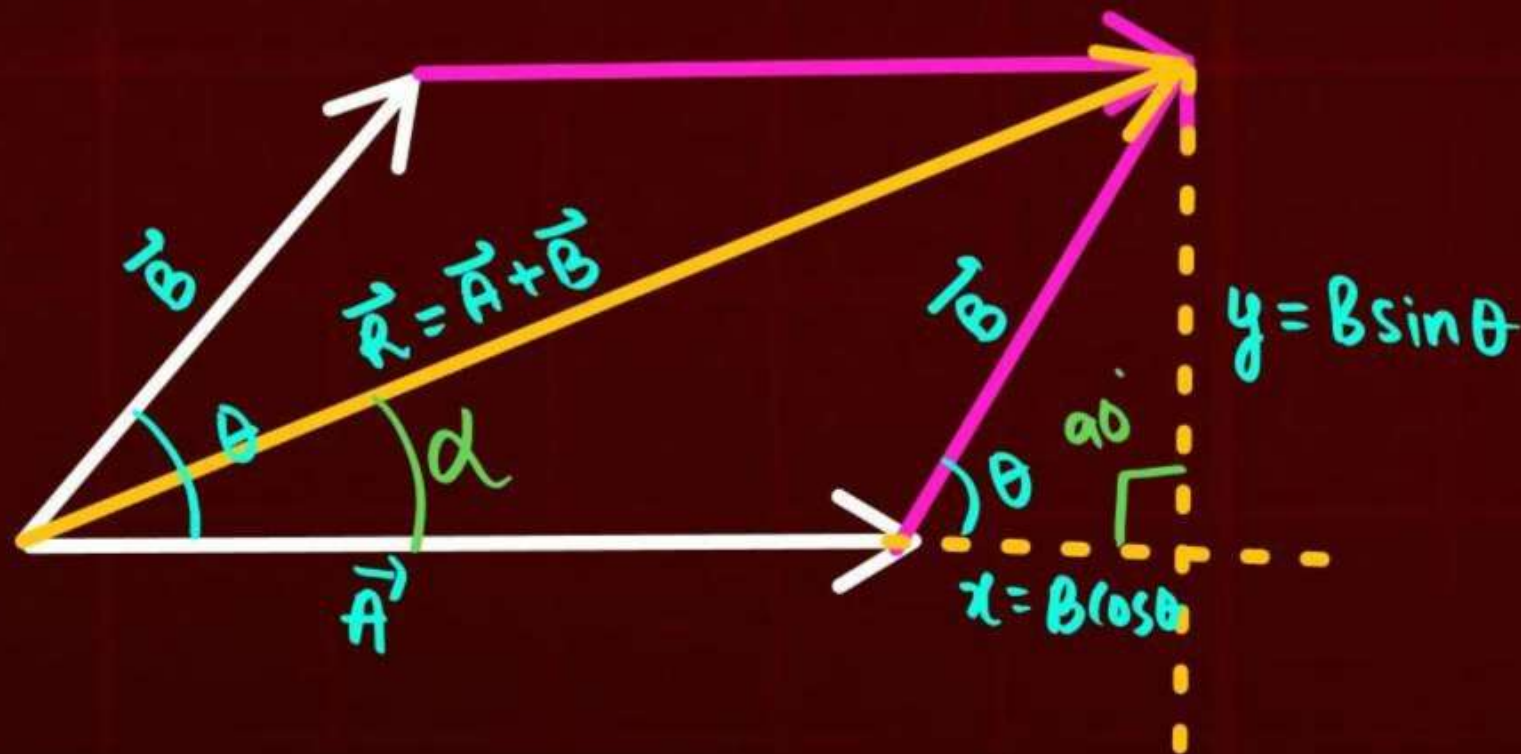


Recap from 2nd Lecture

Parallelogram law of vector Addition

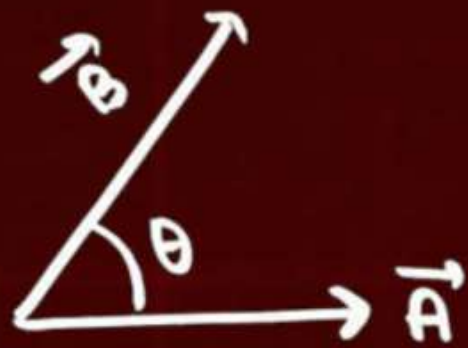
$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$

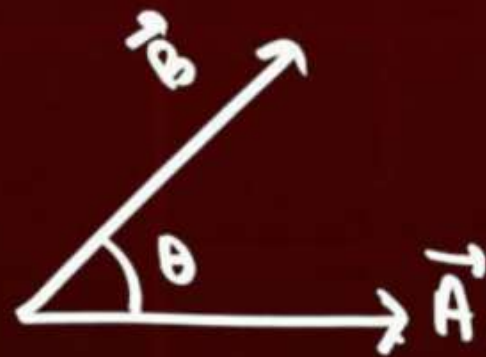


Recap - ③

Cross Product.



$$\vec{B} \times \vec{A} = \underbrace{BA \sin \theta}_{m} \hat{n}_i$$



$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}_{\underline{0}}$$

Properties of Cross Product.

- ① It is a vector product.
- ② Direction is given by **Screw Rule**
- ③ $AB \sin 90^\circ = AB \times 1 = \underline{AB}$ (max)
- ④ $AB \sin 0 = AB \times 0 = 0$ (min)
- ⑤ $\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$ do not show commutative property



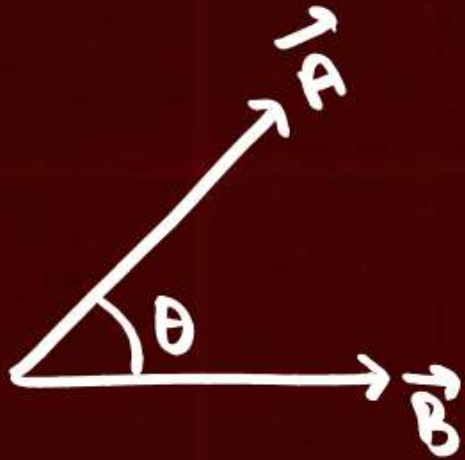
⑥ Distributive property

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

Recap from 4th Lecture

Dot product of two vector.

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$



$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\vec{B} \cdot \vec{A} = BA \cos \theta$$

Properties of Dot product.

- ① $\vec{A} \cdot \vec{B} = AB \cos \theta$ Scalar product.
- ② no direction.
- ③ $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ show commutative p
- ④ $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$
- ⑤ $\vec{A} \cdot \vec{B} = AB \cos \theta$
 $= AB \cos 0 = AB \times 1 = AB$
- ⑥ $\vec{A} \cdot \vec{B} = AB \cos \theta$
 $= AB \cos 90^\circ = AB \times 0 = 0$
- ⑦ $\vec{A} \cdot \vec{B} = AB \cos \theta = AB \cos 180^\circ$
 $= AB \times -1 = -AB$

① Motion in 1D (Straight Line Motion)

lecture no. 1

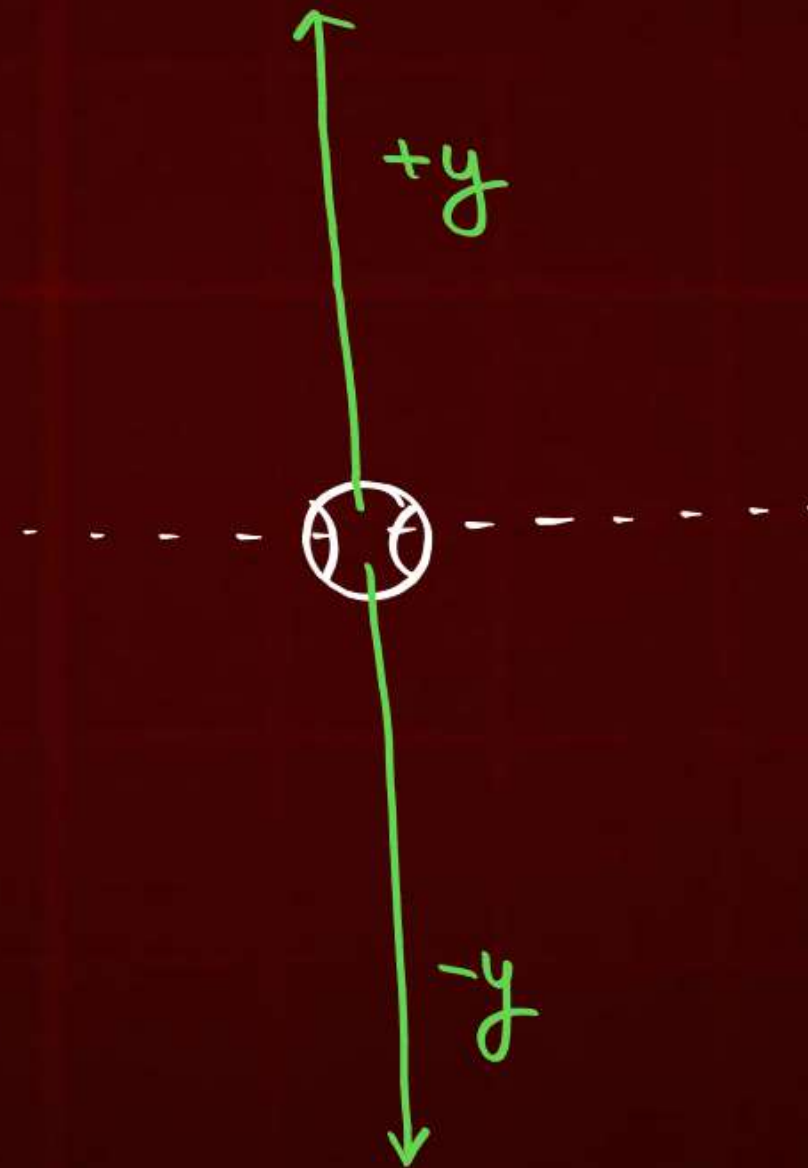
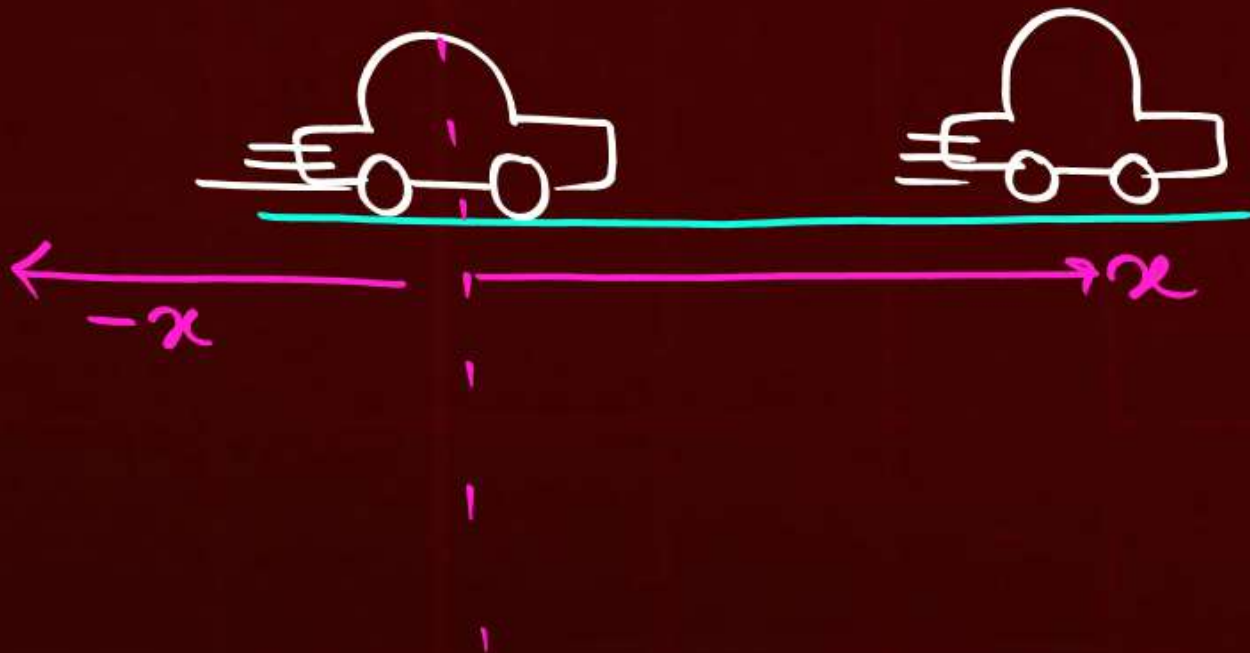


Motion in a straight line

Body moves along the straight line in x -axis or y -axis

example

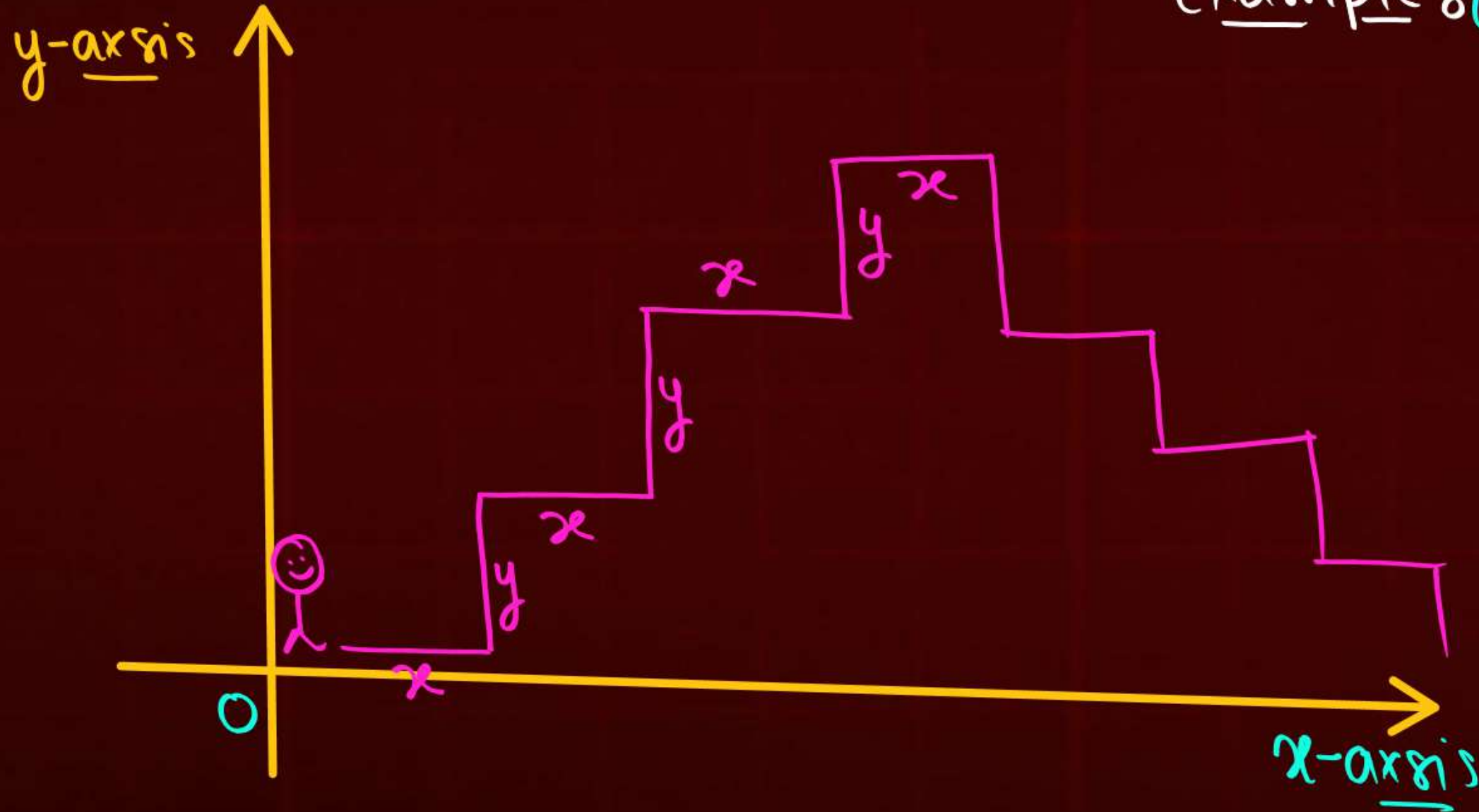
①



② Motion in 2D (Motion in a Plane)

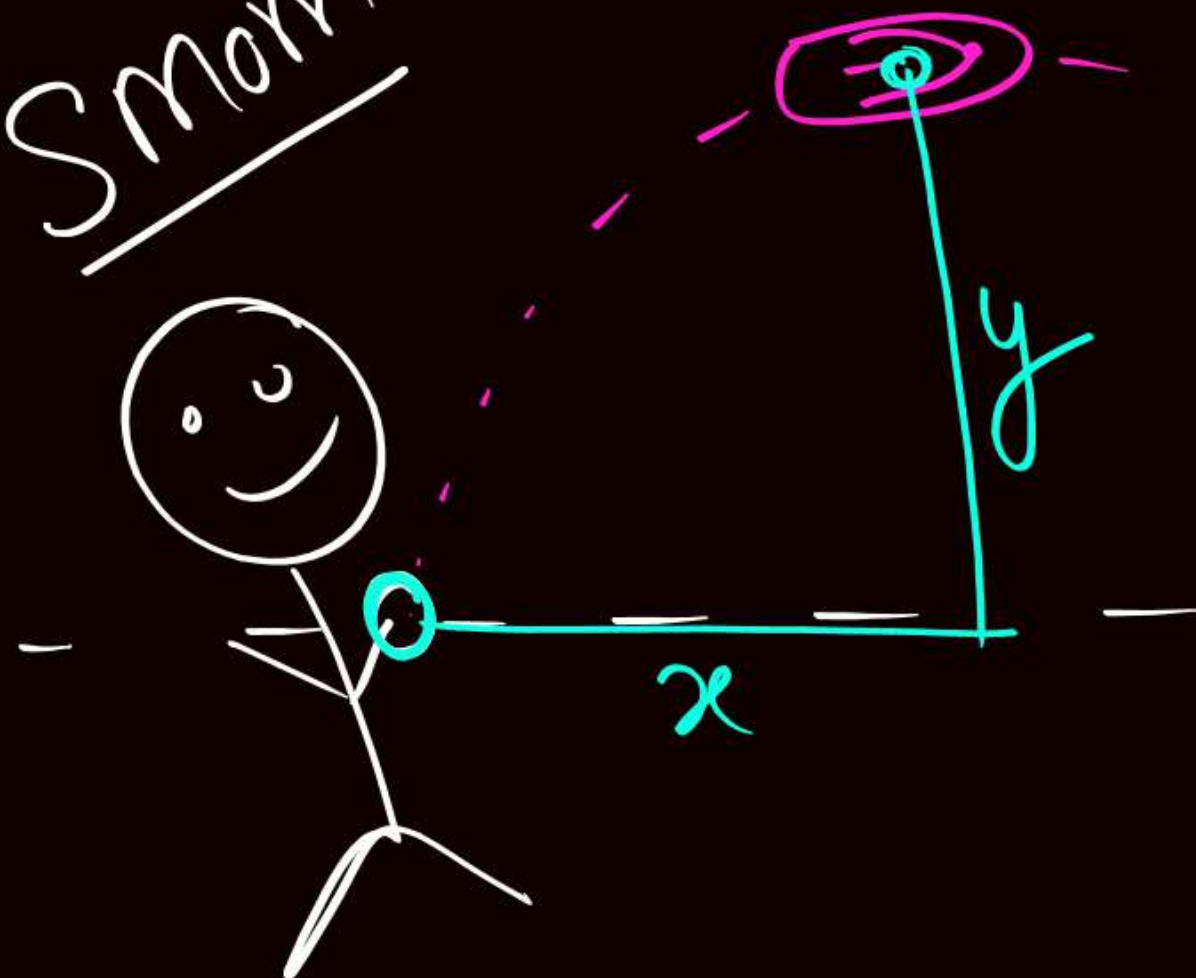
When the Body moves Along x-axis as well as along y-axis

Example of Projectile motion



2025
2026
2024
2023
2019
2017

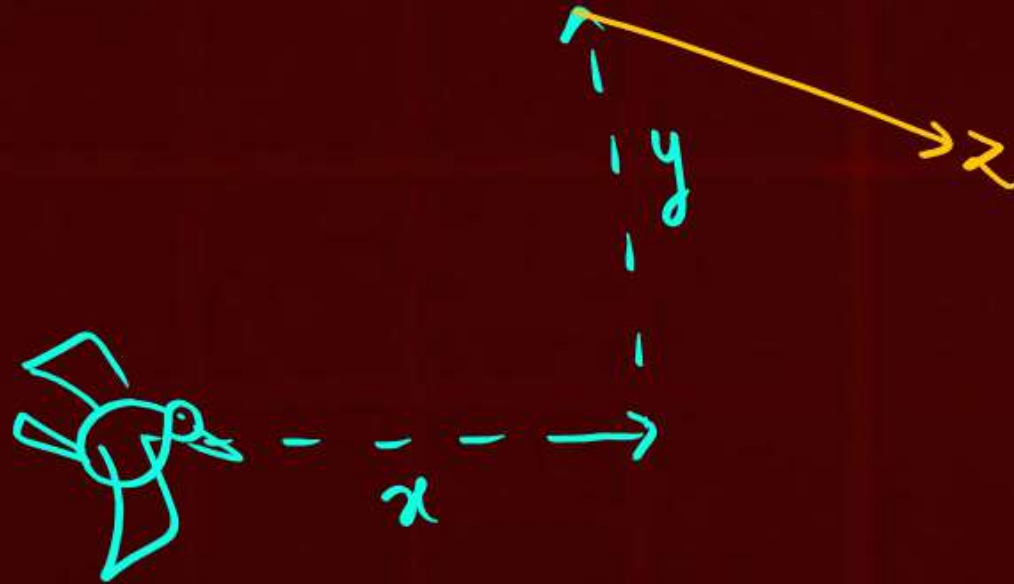
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② Motion in 3D (Motion in a Plane)

When Body use three axis to move

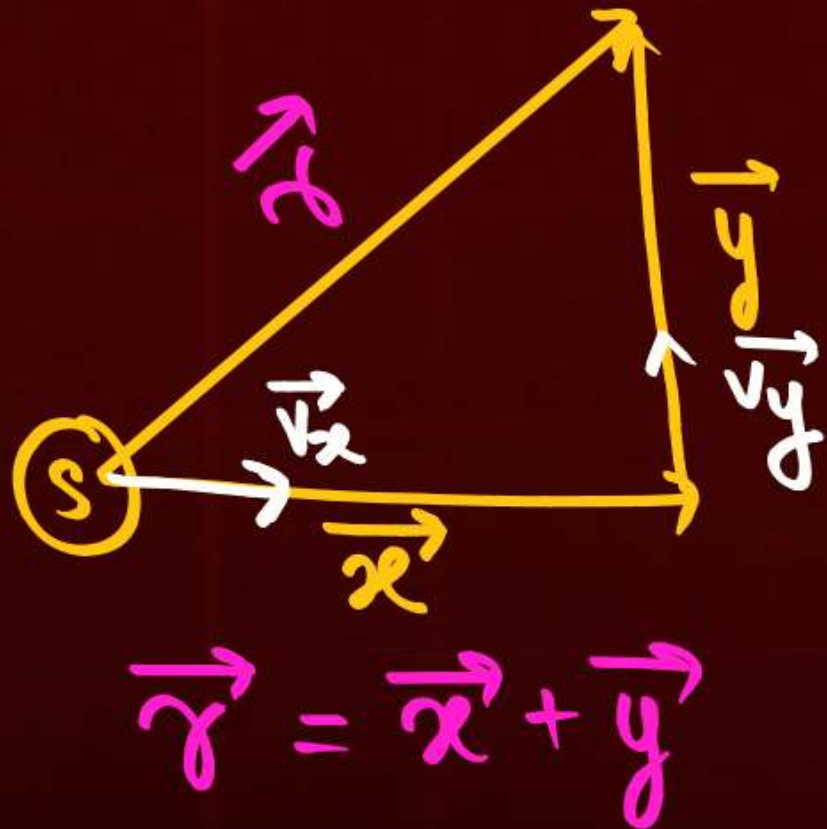
Example: Bird Fly



Motion in 2D

① Displacement (m) ($\vec{r}$) in 2D

Shortest path
travel by the Body



② Velocity in 2D (m/s)

The rate of change of displacement
is Mean velocity

$$\vec{r} = \vec{x} + \vec{y}$$

on differentiating b/s

$$\frac{d\vec{r}}{dt} = \frac{d\vec{x}}{dt} + \frac{d\vec{y}}{dt}$$

$$\vec{V} = \vec{V}_x + \vec{V}_y$$

③ Acceleration in 2D (m/s²)

Rate of change of velocity

$$\vec{V} = \vec{V}_x + \vec{V}_y$$

$$\frac{d\vec{V}}{dt} = \frac{d\vec{V}_x}{dt} + \frac{d\vec{V}_y}{dt}$$

$$\vec{a} = \vec{a}_x + \vec{a}_y$$

on differentiation b/s

Equation of Motion

2D - motion

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③ Equation of motion

$$\textcircled{1} \underline{v} = u + at$$

$$v_x = u_x + a_x t$$

$$v_y = u_y + a_y t$$

$$\textcircled{2} S = ut + \frac{1}{2}at^2$$

$$S_x = u_x t + \frac{1}{2}a_x t^2$$

$\downarrow$
 x

$$S_y = u_y t + \frac{1}{2}a_y t^2$$

$\downarrow$
 y

$$\textcircled{3} v^2 = u^2 + 2as$$

$$v_x^2 = u_x^2 + 2a_x \underline{S_x}$$

x

$$v_y^2 = u_y^2 + 2a_y \underline{S_y}$$

$\downarrow$
 y

Projectile Motion

(5-mark) 2026, 2025, 2024, 2023, 2022



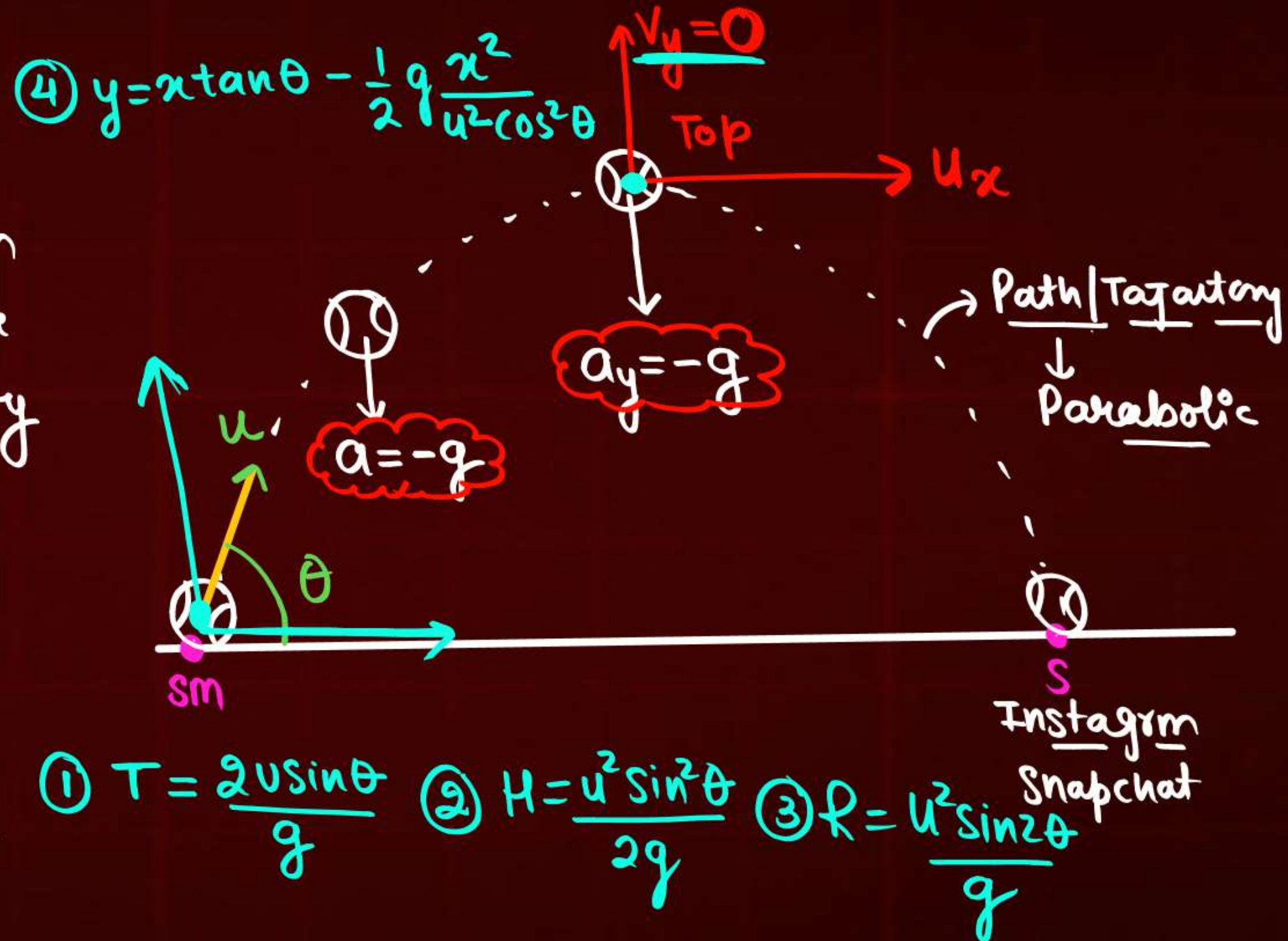
- Example of 2D
- To explain the 2D motion
- when an object is thrown with some Angle under the influence of gravity

$a_y = -g = -9.8 = -10 \text{ m/s}^2$

$a_x = 0$

$v_y = 0$

Path | Tangential → Parabolic



Repeat

$$\textcircled{1} T = \frac{2u \sin \theta}{g}$$

$$\textcircled{2} H = \frac{u^2 \sin^2 \theta}{2g}$$

$$\textcircled{3} R = \frac{u^2 \sin 2\theta}{g}$$

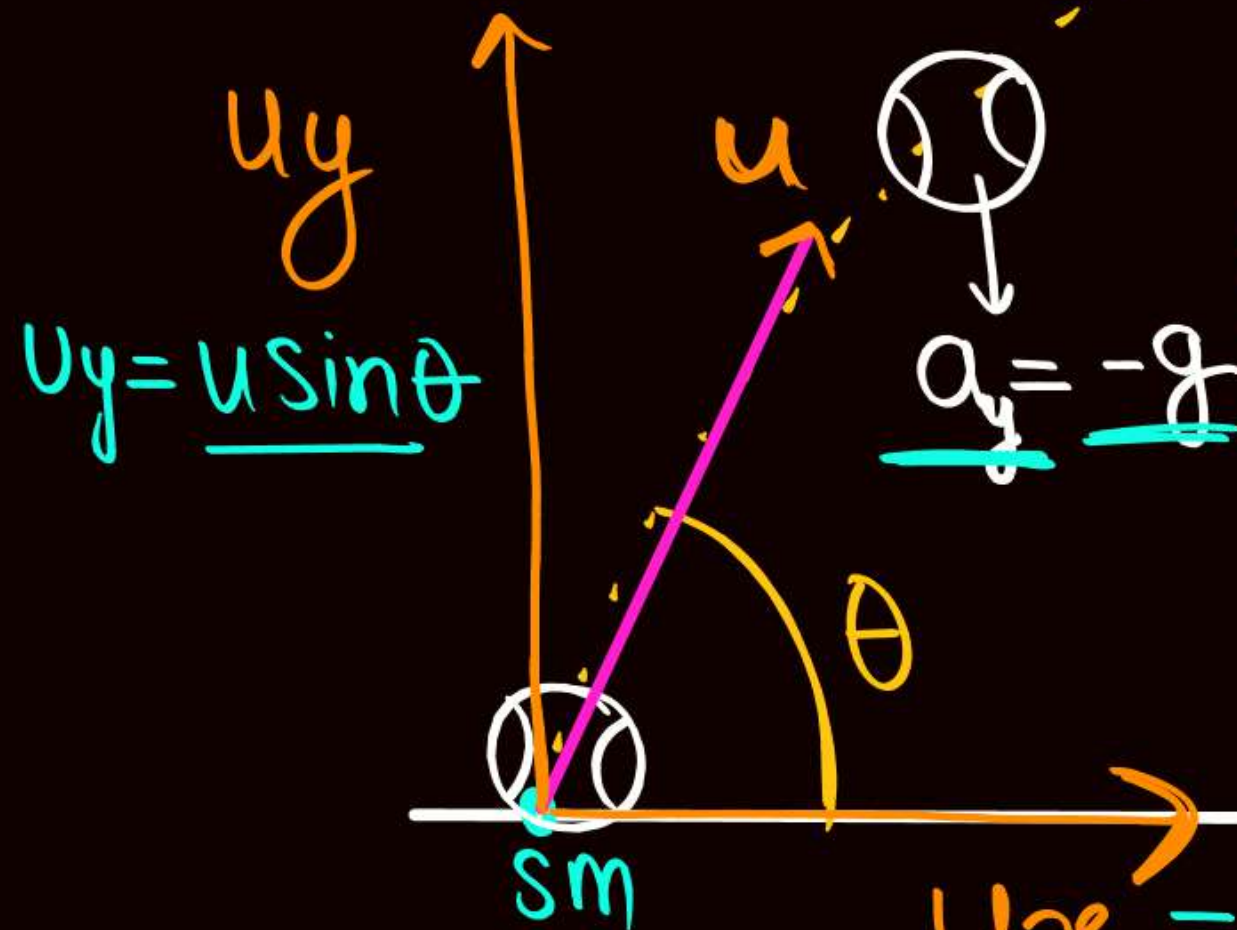
$$v_y = 0$$



Top

$$u_x = u \cos \theta$$

path / Trajectory
(parabolic)



$$\textcircled{4} y = x \tan \theta - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$



s

$$u_x = u \cos \theta$$

Projectile Motion

1. Time of flight

Projectile Motion

2. Maximum Height

Projectile Motion

3. Range

SUMMARY



Displacement 2D

Velocity 2D

acc $\rightarrow$ 2D

Equation of Motion

Projectile Motion

HOMework



photo → projectile motion

Thank
You